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🚀 Preparing for a **#VLSI #Digital** Design interview involves a strong understanding of **#Sequential** circuits. Below is a refined and well-structured list of few important topics and questions related to Sequential circuits that are highly relevant for VLSI interviews:

#Important Topics on **#Sequential_Circuits** in Digital **#Design_for_VLSI**

1. Truth table and excitation table for JK, SR, D, and T flip-flops.
2. Conversion between flip-flops (e.g., JK to SR, SR to D and T, etc.).
3. Implement flip-flop using 2x1 multiplexers.
4. Difference between latches and flip-flops, including use cases for each.
5. Problems in SR flip-flops and how to resolve them.

6. Race-around condition in JK flip-flops and solutions.
7. Master-slave JK flip-flop configuration.
8. What is Clock Domain Crossing (#CDC) and how to avoid CDC issues.
9. Metastability and how to prevent it.
10. Mean Time Between Failure (#MTBF) and its significance.
11. Use of synchronizers in digital circuits; difference between 2-FF and 3-FF synchronizers.
12. Types of synchronous and asynchronous #counters.
13. Use of up/down counters.
14. Design a #clock_divider circuit for different frequencies and duty cycles.
 - ✦ Divide-by-2 Clock Divider (50% Duty Cycle)
 - ✦ Divide-by-3 Clock Divider (33% Duty Cycle)
 - ✦ Divide-by-5 Clock Divider (Custom Duty Cycle)
 - ✦ Divide-by-N Clock Divider (Custom Duty Cycle)
 - ✦ Clock Divider with 75% Duty Cycle for fclk/3
 - ✦ Clock Divider with 25% Duty Cycle fclk/3
15. Design a #frequency doubler circuit.
16. Block diagram of synchronous and asynchronous #FIFO; explain empty/full conditions.
17. Difference between #Mealy and #Moore state machines.
18. Choosing between Mealy and Moore machines in design.
19. Mod counter design with specific duty cycle.
20. Sequence detector #FSM design for "1010" (overlapping and non-overlapping).
21. Types of shift registers: #SISO, #SIPO, #PIPO, #PISO (#universal shift register).
22. #BCD counter design and implementation.
23. FSM implementation for double sequence detection (e.g., 1010 and 1101).
24. What is a transparent #latch?
25. Conversion between different number systems.
26. Why Gray counters are preferred over binary counters in digital circuits.
27. Difference between synchronous and asynchronous reset.
28. Design of synchronous and asynchronous D flip-flops.
29. Implement division by 2^n circuit.
30. Implement multiplication by 2^n circuit.
31. Design a circuit to compute $3*a$ without using shifters.
32. Types of #hazards: static 0, static 1, dynamic hazard, and time-zero race condition.
33. Time-borrowing concept in digital design.
34. Why is #JK flip-flop named JK? Similarly for #SR, #D, and #T flip-flops.
35. Practice implementing circuits based on #output #waveforms.

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